

IT PROFESSIONAL PORTFOLIO

Cloud Engineering and Security Projects

PROJECT 41

Create a Forensic EBS Snapshot and Share It with SecurityAudit

Amazon EBS | Snapshots | EC2 | Cross-Account Sharing | Forensics

AUTHOR	Michael Salazar
ENVIRONMENT	Personal AWS Lab
VERSION	v1.0.0
RELEASE DATE	July 30, 2026
STATUS	Part 1 Complete

Executive Summary

Created a **CSI** CloudFormation stack from the provided template as part one of a two-part forensic snapshot lab. In EC2, identified the storage volume attached to the **Hacked** instance, created an Amazon EBS snapshot named **CSI-Incident-1**, waited for the snapshot to complete, and shared the snapshot privately with the **SecurityAudit** account for cross-account forensic analysis in the next lab.

FORENSIC PATTERN

This is a forensic acquisition pattern: copy the suspected disk to a snapshot, preserve the original volume, then share the immutable copy with the security account for later analysis.

Business Problem

During an incident, responders often need to preserve evidence from a compromised instance before deeper analysis. In AWS, the relevant disk evidence usually lives on EBS volumes. EBS snapshots provide a point-in-time copy that can be shared to a security account so analysis can occur away from the workload account.

The same capability has security risk: snapshot sharing is also a data-exfiltration technique when permissions are weak. This lab documents both the legitimate forensic workflow and the control risk around snapshot sharing.

Objectives

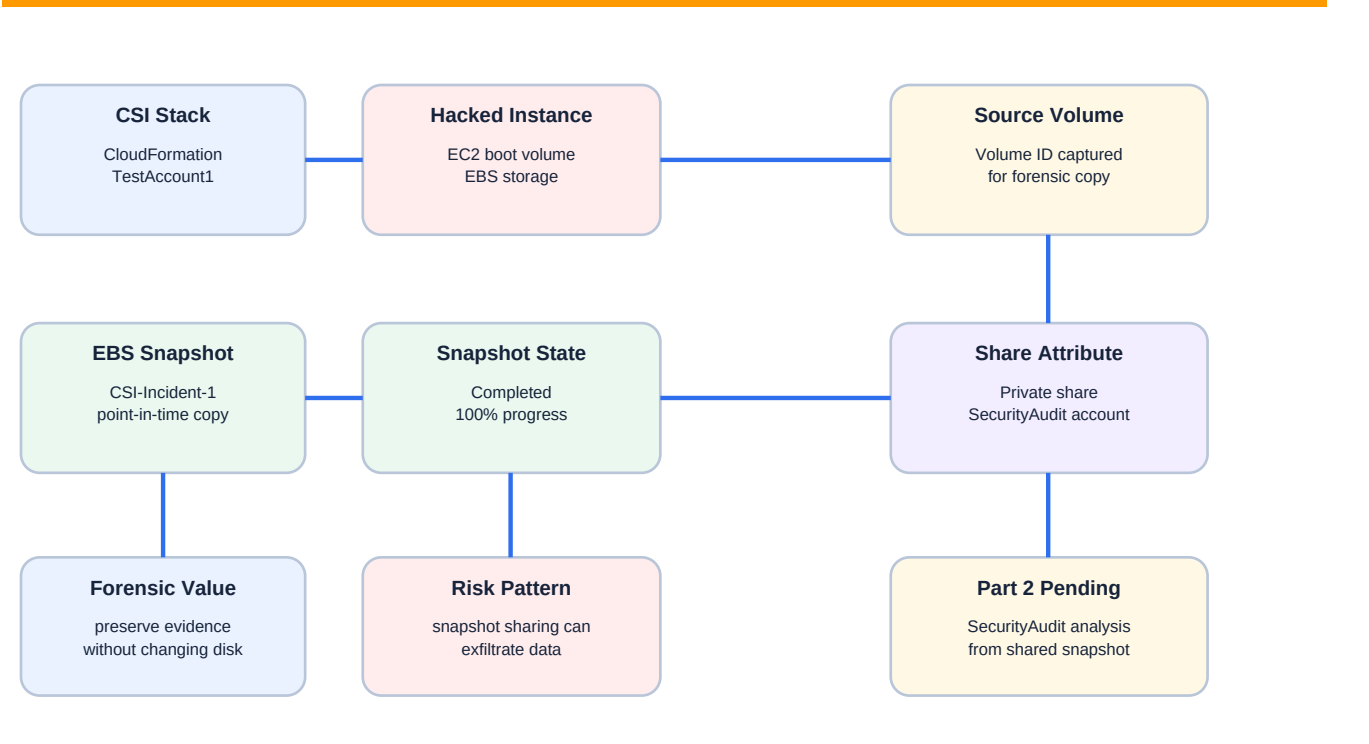
- * Create the CSI CloudFormation stack from the provided lab template.
- * Locate the Hacked EC2 instance created by the stack.
- * Identify the EBS boot volume attached to the Hacked instance.
- * Create a point-in-time EBS snapshot of that source volume.
- * Name the snapshot CSI-Incident-1 for incident tracking.
- * Verify snapshot completion before sharing.
- * Modify snapshot sharing permissions to share the snapshot with the SecurityAudit account.
- * Keep the snapshot private, not public.
- * Prepare the snapshot for Part 2 forensic analysis in SecurityAudit.

Environment

Cloud Provider	Amazon Web Services (AWS)
Primary Services	Amazon EC2, Amazon EBS, EBS Snapshots, AWS CloudFormation, AWS Organizations
Account Context	TestAccount1 / AdministratorAccess
Stack Name	CSI

Template Source	Provided CloudFormation lab template
Compromised Instance Label	Hacked
Evidence Source	EBS boot volume attached to Hacked instance
Snapshot Description	CSI-Incident-1
Snapshot Size Evidence	1.64 GiB full snapshot size, 8 GiB source volume size
Snapshot State	Completed / 100% progress
Snapshot Encryption	Not encrypted in lab evidence
Share Target	SecurityAudit account
Share Type	Private cross-account snapshot share
Project State	Part 1 complete - forensic analysis continues in Part 2

Architecture



Snapshot acquisition preserves a point-in-time copy and transfers evidence to a security account for analysis.

Figure 1 - Forensic snapshot acquisition and cross-account sharing workflow.

Implementation Summary

1. Signed into TestAccount1 with AdministratorAccess.
2. Created the CSI CloudFormation stack from the provided template.
3. Opened EC2 and located the Hacked instance created by the stack.
4. Opened the Storage tab for Hacked to identify the attached EBS volume ID.
5. Selected the EBS volume and started Create snapshot.
6. Entered CSI-Incident-1 as the snapshot description.
7. Created the snapshot and waited until snapshot status was Completed / Available.
8. Opened snapshot details and reviewed source volume, size, completion state, and encryption status.
9. Modified snapshot sharing permissions.
10. Kept sharing private and added the SecurityAudit account as the allowed shared account.
11. Confirmed the shared accounts section showed one shared account.
12. Preserved the snapshot for the next lab, where the shared forensic copy will be analyzed.

Forensic Control Map

EBS volume identification	Hacked instance storage volume ID captured	Ensures the snapshot is taken from the correct suspected disk.
Snapshot description	CSI-Incident-1	Creates a traceable incident label for the forensic copy.
Snapshot immutability	Snapshot cannot be edited in place	Preserves a point-in-time evidence copy for analysis.
Private sharing	Shared only with SecurityAudit account	Moves evidence to a security environment without making it public.
Security risk note	ModifySnapshotAttribute can share snapshots	Same capability can be abused for exfiltration if permissions are too broad.
Part 2 handoff	Snapshot retained for SecurityAudit analysis	Supports future investigation without modifying the original instance volume.

Evidence

Figure 2 - Snapshot Creation from Hacked Instance Volume

Create snapshot info

Create a point-in-time snapshot to back up the data on an Amazon EBS volume to Amazon S3.

Source volume

Volume ID

Availability Zone

usw2-az2 (us-west-2a)

Snapshot details

Description

Add a description for your snapshot

CSI-Incident-1

255 characters maximum

Encryption info

Not encrypted

Tags info

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

No tags associated with the resource.

Add new tag

You can add up to 50 tags.

Cancel

Create snapshot

The snapshot creation evidence shows the source volume selected, the snapshot description set to CSI-Incident-1, Availability Zone usw2-az2 / us-west-2a, and encryption listed as Not encrypted. The source volume ID is redacted.

Figure 3 - Completed Snapshot Shared with SecurityAudit

Details

Snapshot ID

Owner

Description

CSI-Incident-1

Source volume

Volume ID

Volume size

8 GiB

Full snapshot size

1.64 GiB

Started

Thu Jul 30 2026 11:27:26 GMT-0600 (Mountain Daylight Time)

Progress

100%

Product codes

Snapshot status

Completed

Fast snapshot restore

Encryption

Not encrypted

KMS key ID

KMS key alias

KMS key ARN

Snapshot settings

Storage tier

Tags

Snapshot Lock

Lock mode

Not locked

Share permissions

Snapshot share permissions

Private

The snapshot is shared only with AWS accounts that you specified.

Modify permissions

Shared accounts (1)

The snapshot is shared with the following AWS accounts.

Find shared accounts

AWS account ID

Add account ID

The shared snapshot evidence shows description CSI-Incident-1, Progress 100%, Snapshot status Completed, full snapshot size 1.64 GiB, source volume size 8 GiB, encryption Not encrypted, private snapshot share permissions, and Shared accounts (1). Snapshot ID, owner, source volume, and account ID are redacted.

Security Analysis

EBS snapshots are valuable for incident response because they create a point-in-time disk copy without changing the source volume under investigation.

Cross-account sharing moves the forensic copy into a security account where responders can analyze evidence without granting broad access inside the workload account.

Snapshot sharing is also a common data-exfiltration path. Anyone with permission to modify snapshot attributes can potentially share a copy of sensitive storage with another AWS account.

The lab snapshot is unencrypted. This simplifies sharing for the lab, but production forensic workflows should define KMS key ownership, cross-account key permissions, and encryption requirements.

The snapshot was shared privately, not publicly. Public snapshot exposure would be a major data-loss risk.

For a real incident, responders should also preserve CloudTrail, IAM context, GuardDuty/Security Hub findings, network evidence, instance tags, and chain-of-custody records.

Lessons Learned

EC2 instances use EBS volumes as virtual disks, and the boot volume is the primary source for disk-level investigation.

An EBS snapshot is a point-in-time copy of an EBS volume and can be used for backup, migration, recovery, or forensic analysis.

Snapshots cannot be edited in place, which gives them some forensic value similar to a write-blocked evidence copy.

Snapshot permissions are modified directly on the snapshot as share attributes, which differs from normal IAM or resource-policy sharing patterns.

The same snapshot sharing mechanism supports legitimate security investigations and attacker data exfiltration.

Part 1 of the workflow is acquisition and sharing; Part 2 will analyze the shared copy from the SecurityAudit side.

Production Improvements

- * Use a dedicated IncidentResponse or Forensics account instead of reusing SecurityAudit.
- * Automate snapshot acquisition with incident-response runbooks or Lambda/Step Functions.
- * Tag snapshots with incident ID, source account, source instance, source volume, responder, timestamp, and retention requirement.
- * Require encryption for forensic snapshots and preconfigure KMS grants for the forensic account.
- * Block public snapshot sharing at the organization/account level.
- * Restrict `ec2:ModifySnapshotAttribute` and `ec2:CreateSnapshot` with least privilege and approval workflows.
- * Monitor snapshot creation and sharing with CloudTrail, EventBridge, Security Hub, and AWS Config.
- * Use Snapshot Lock or retention controls when legal hold or evidence preservation requirements apply.
- * Document chain of custody before deleting source infrastructure or forensic copies.

Skills Demonstrated

EBS volume identification	EBS snapshots	Snapshot sharing
Incident evidence handling	SecurityAudit handoff	CloudFormation lab setup
Exfiltration risk analysis	Private permissions	Cost/retention awareness

Resume Bullet

RESUME	Created a forensic Amazon EBS snapshot from a suspected compromised EC2 instance, labeled the evidence copy as CSI-Incident-1, validated snapshot completion, and shared the private snapshot with a SecurityAudit account for cross-account forensic analysis while documenting snapshot-sharing exfiltration risks.
--------	---

STAR Interview Story

Situation: A lab incident scenario required preserving evidence from a Hacked EC2 instance before analysis.

Task: Identify the instance storage volume, create a forensic snapshot, and share it with the security account for investigation.

Action: I created the CSI stack, located the Hacked instance storage volume, created the CSI-Incident-1 EBS snapshot, verified completion, and modified the snapshot permissions to share it privately with SecurityAudit.

Result: The forensic acquisition was completed without modifying the source volume, and the snapshot was staged for cross-account investigation in Part 2.

Version History

v1.0.0	July 30, 2026	Initial documented implementation. Created CSI stack, identified the Hacked instance EBS volume, created CSI-Incident-1 forensic snapshot, verified completion, shared the private snapshot with SecurityAudit, documented snapshot exfiltration risk, and embedded provided evidence using portfolio documentation standard v1.0.1.

References

- * Cloud Security Lab a Week (S.L.A.W.) - Create a (Forensic) Snapshot and Play CSI!.
- * User-provided snapshot creation evidence screenshot.
- * User-provided shared snapshot evidence screenshot.
- * Project 40 - Simulate Cryptomining User Data Abuse and Validate GuardDuty Detection.